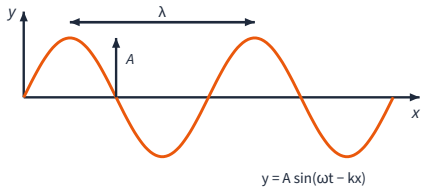
CHAPTER 21 WAVE MOTION

Key idea: a progressive wave carries energy without carrying matter; overlapping waves add.

1 PROGRESSIVE WAVES



$$v = f\lambda$$

$$y = A\sin(\omega t - kx)$$

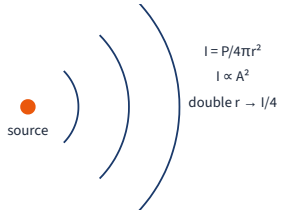
- $k = \dfrac{2\pi}{\lambda}, v = \dfrac{\omega}{k}$
- $+kx$: travels in the $-x$ direction
- Transverse: $\perp$ to travel; longitudinal: $\parallel$

2 PHASE DIFFERENCE

$$\phi = \frac{2\pi}{\lambda}\Delta x \qquad \phi = \frac{2\pi}{T}\Delta t$$

In phase: $\Delta x = n\lambda$. Antiphase: $\Delta x = (n + \frac{1}{2})\lambda$.

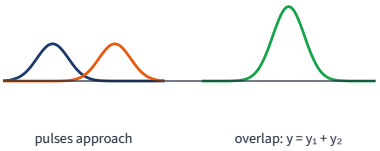
3 INTENSITY



$$I = \frac{P}{4\pi r^2} \qquad I \propto A^2$$

Power per unit area, W m^{-2}. Point source: $I \propto 1/r^2$ so $A \propto 1/r$.

4 SUPERPOSITION



Constructive path difference $n\lambda$

Destructive path difference $(n + \frac{1}{2})\lambda$

Coherent: same frequency, constant phase difference.

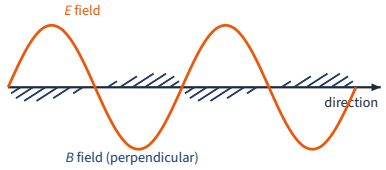
5 STANDING WAVES

$$y = 2A\sin kx\ \cos\omega t$$

- Two equal waves travelling in **opposite** directions
- Node to node $= \lambda/2$; node to antinode $= \lambda/4$
- String fixed both ends: $L = \dfrac{n\lambda}{2}$, $f_n = \dfrac{nv}{2L}$

$$v = \sqrt{\frac{T}{\mu}}$$

6 ELECTROMAGNETIC WAVES



$$c = \frac{1}{\sqrt{\mu_0\varepsilon_0}} \qquad E_0 = cB_0$$

$E \perp B \perp$ direction, in phase. Radio → microwave → IR → visible → UV → X-ray → γ (λ decreasing).

7 PROGRESSIVE VS STANDING

	Progressive	Standing
Energy	transferred	not transferred
Amplitude	same everywhere	zero at nodes, max at antinodes
Phase	changes along the wave	same within a loop; antiphase in next loop
Wave profile	moves	does not move

8 COMMON SECTION A TRAPS

- Node spacing is $\lambda/2$, not λ.
- Reading k as the wavelength; $\lambda = 2\pi/k$.
- Doubling the distance quarters I but halves A.
- μ is mass per unit **length** (kg m^{-1}).

EXAM TIPS

- ✓From $y = A\sin(\omega t - kx)$ read A, ω and k first.
- ✓Draw the harmonics with N and A labelled.
- ✓Use ratios for the inverse square law.
- ✓State the sources are coherent before discussing interference.